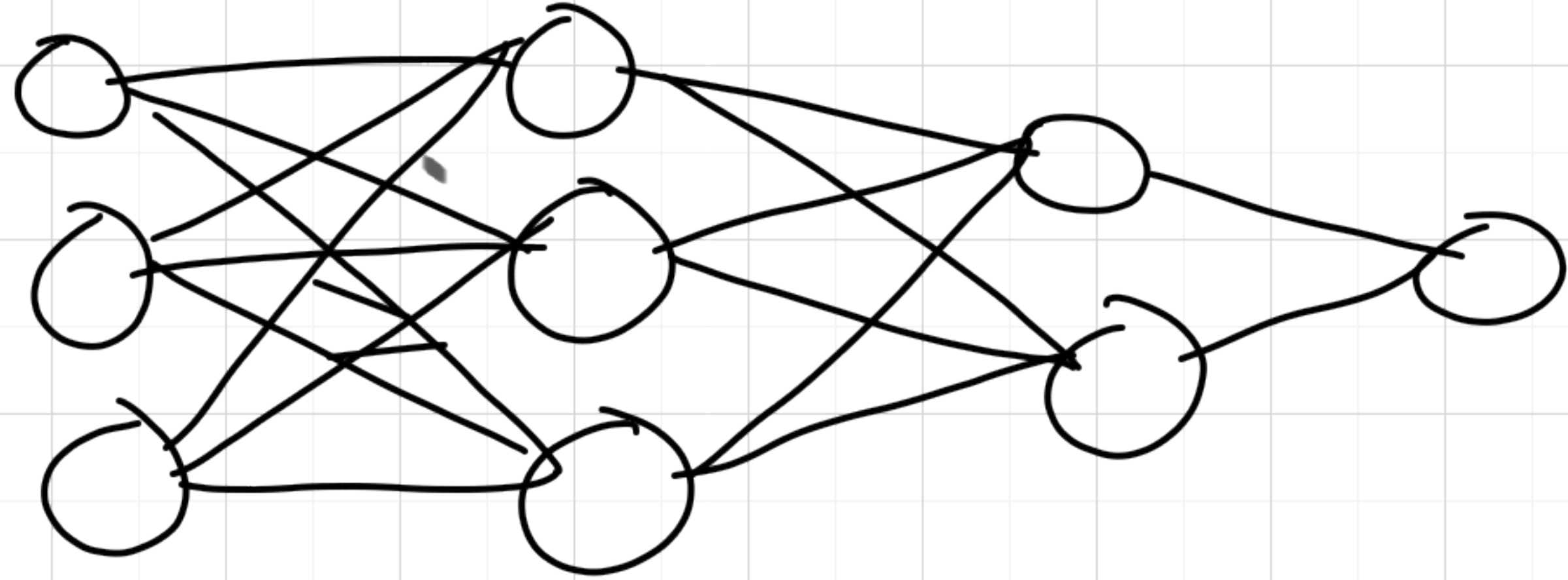
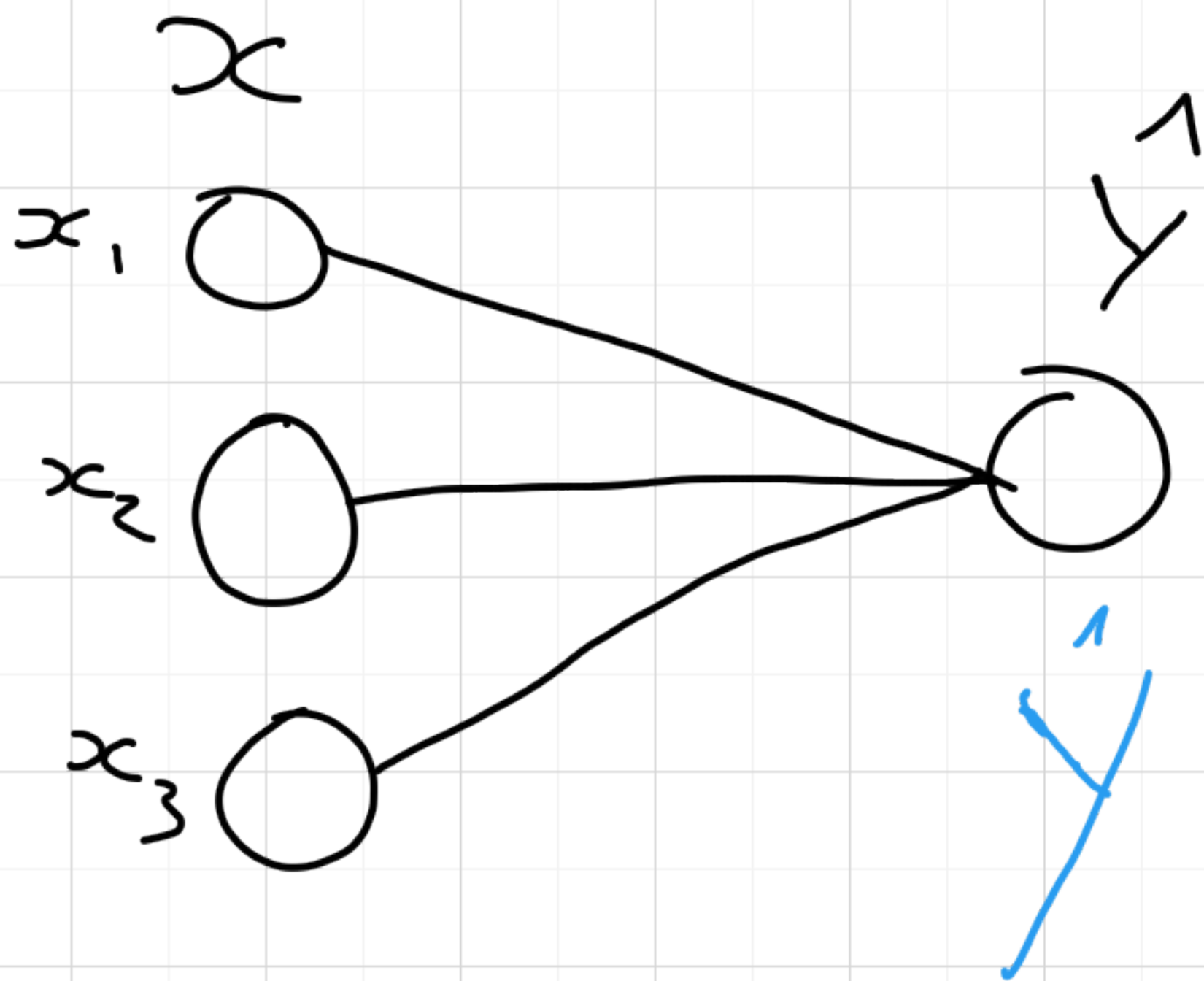


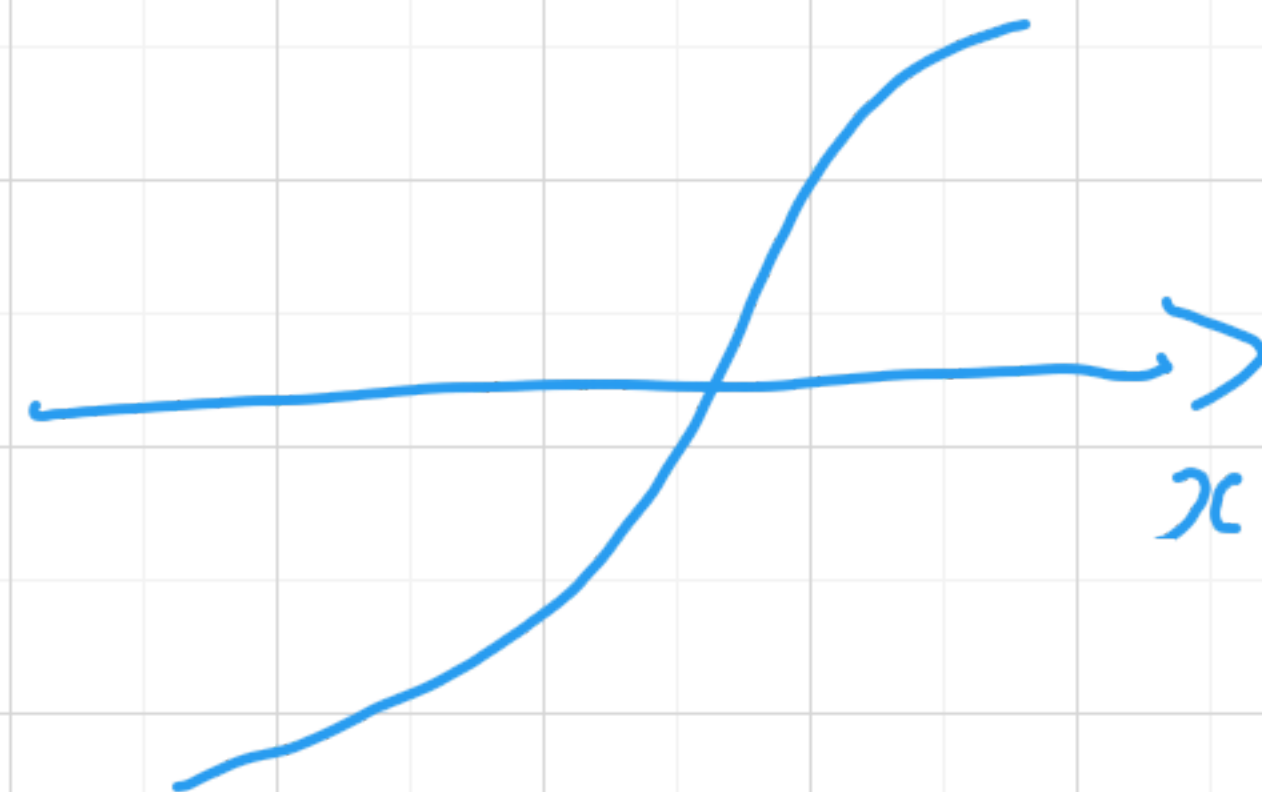
MLP - Multi-layer Perceptron



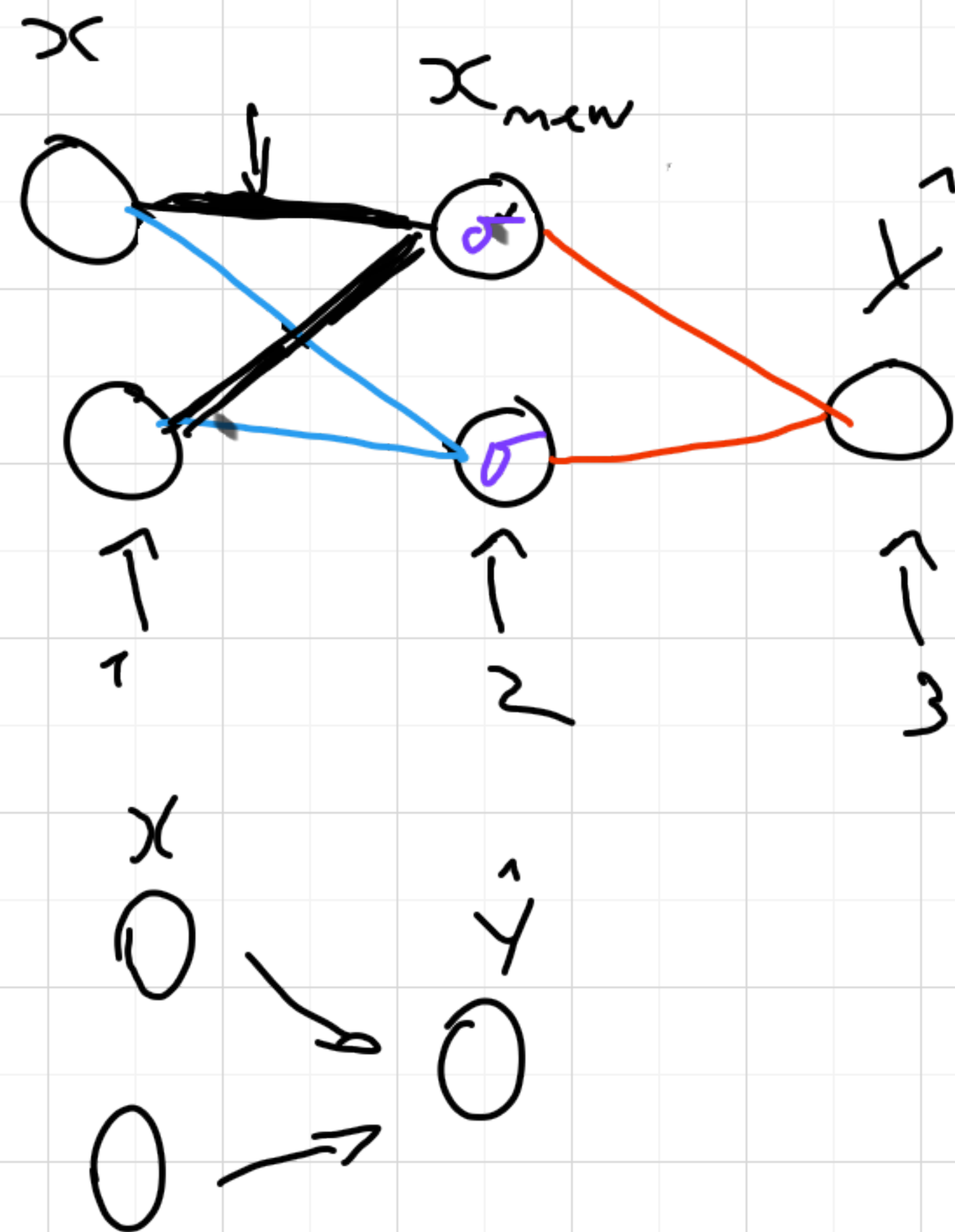


$$\hat{y} = \sigma(w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



New dimensions



$$x_{new1} = \sigma \left(W_1^{1,1} \cdot x_1 + W_2^{1,1} \cdot x_2 + b_0^{1,1} \right)$$

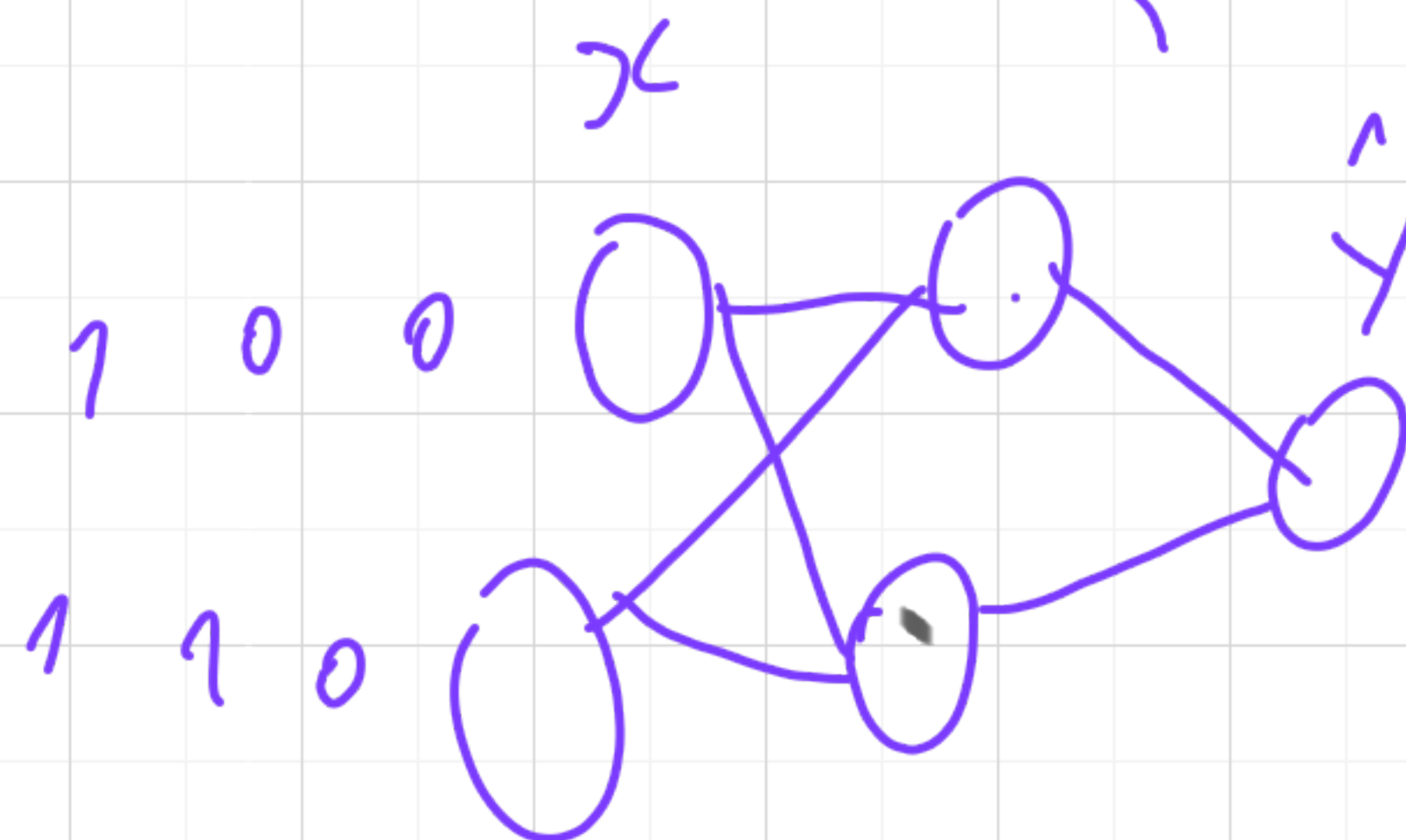
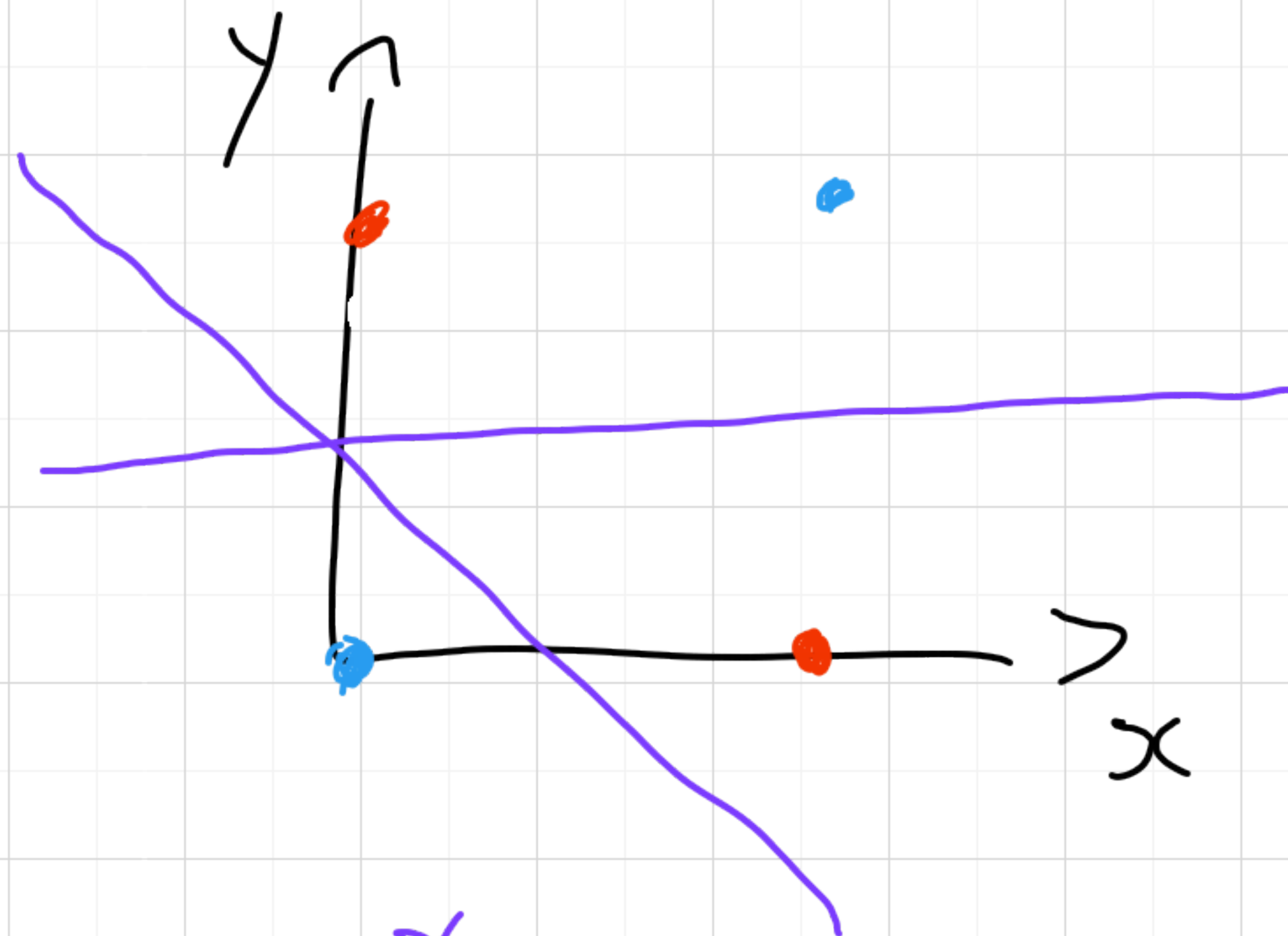
$$x_{new2} = \sigma \left(W_1^{1,2} \cdot x_1 + W_2^{1,2} \cdot x_2 + b_0^{1,2} \right)$$

$$\hat{y} = W_1^{\hat{y}} \cdot x_{new1} + W_2^{\hat{y}} \cdot x_{new2}$$

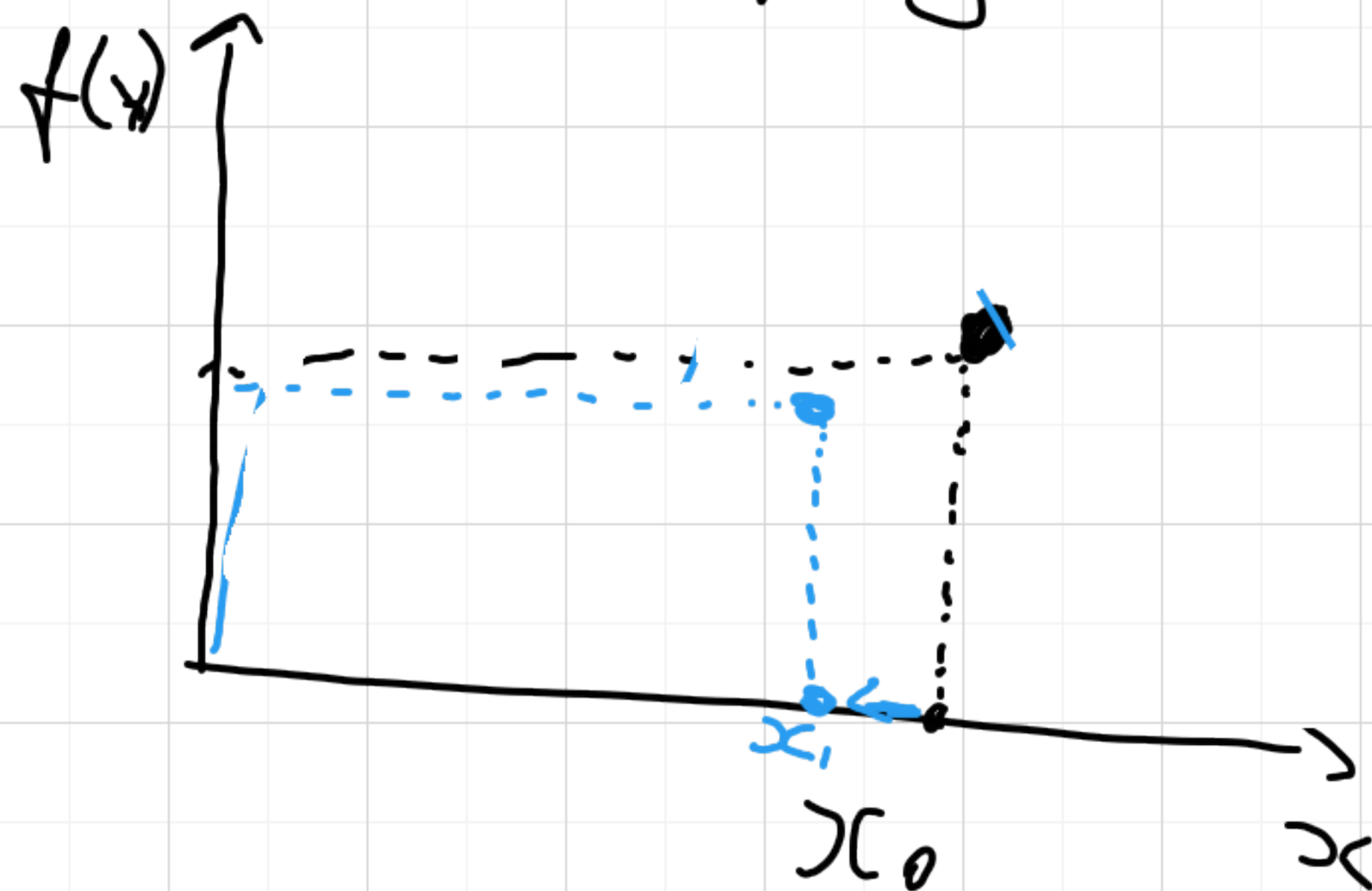
$$\sigma \rightarrow \begin{cases} \text{Sigmoid}(x) \\ \tanh(x) \\ \text{ReLU}(x) \\ \vdots \end{cases}$$

$$\begin{aligned} + b_0^{\hat{y}} &= W^{\hat{y}} \cdot x_{\text{new}} + b_0^{\hat{y}} \\ &= W^{\hat{y}} \left(W^{\text{new}} \cdot x + b_0^{\text{new}} \right) + b_0^{\hat{y}} = W_{\text{new}} \cdot x + \underbrace{W^{\hat{y}} \cdot b_0^{\text{new}} + b_0^{\hat{y}}}_{b_{0\text{new}}} = W \cdot x \end{aligned}$$

XOR	0	1
0	0	1
1	1	0

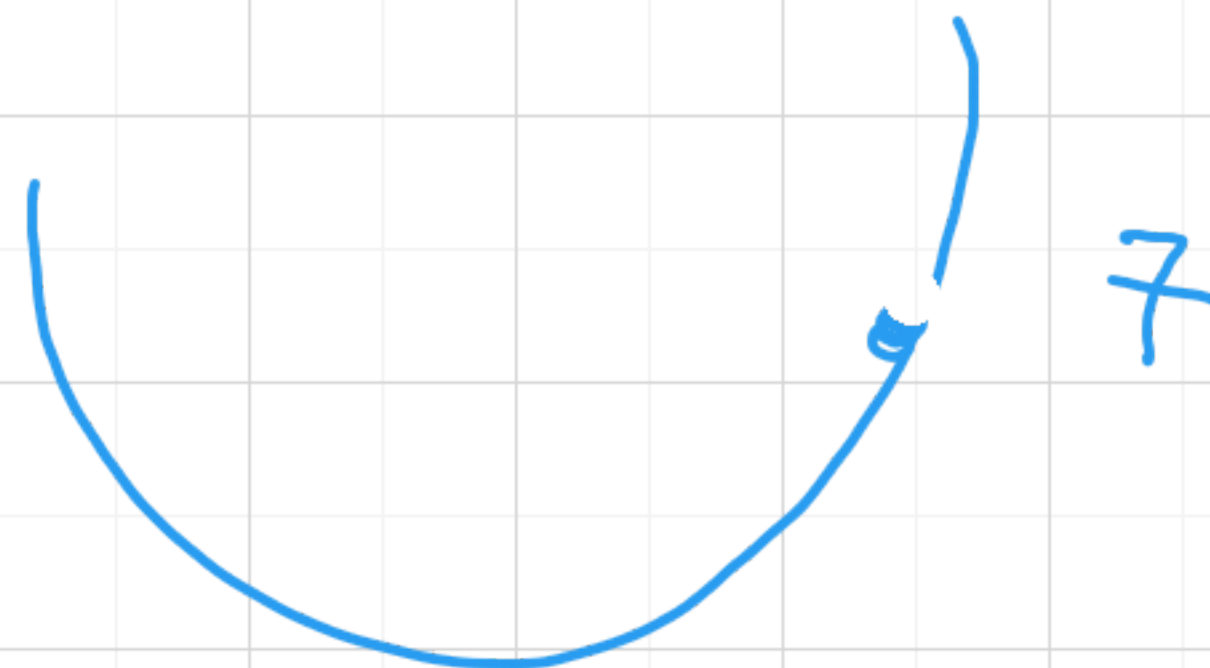


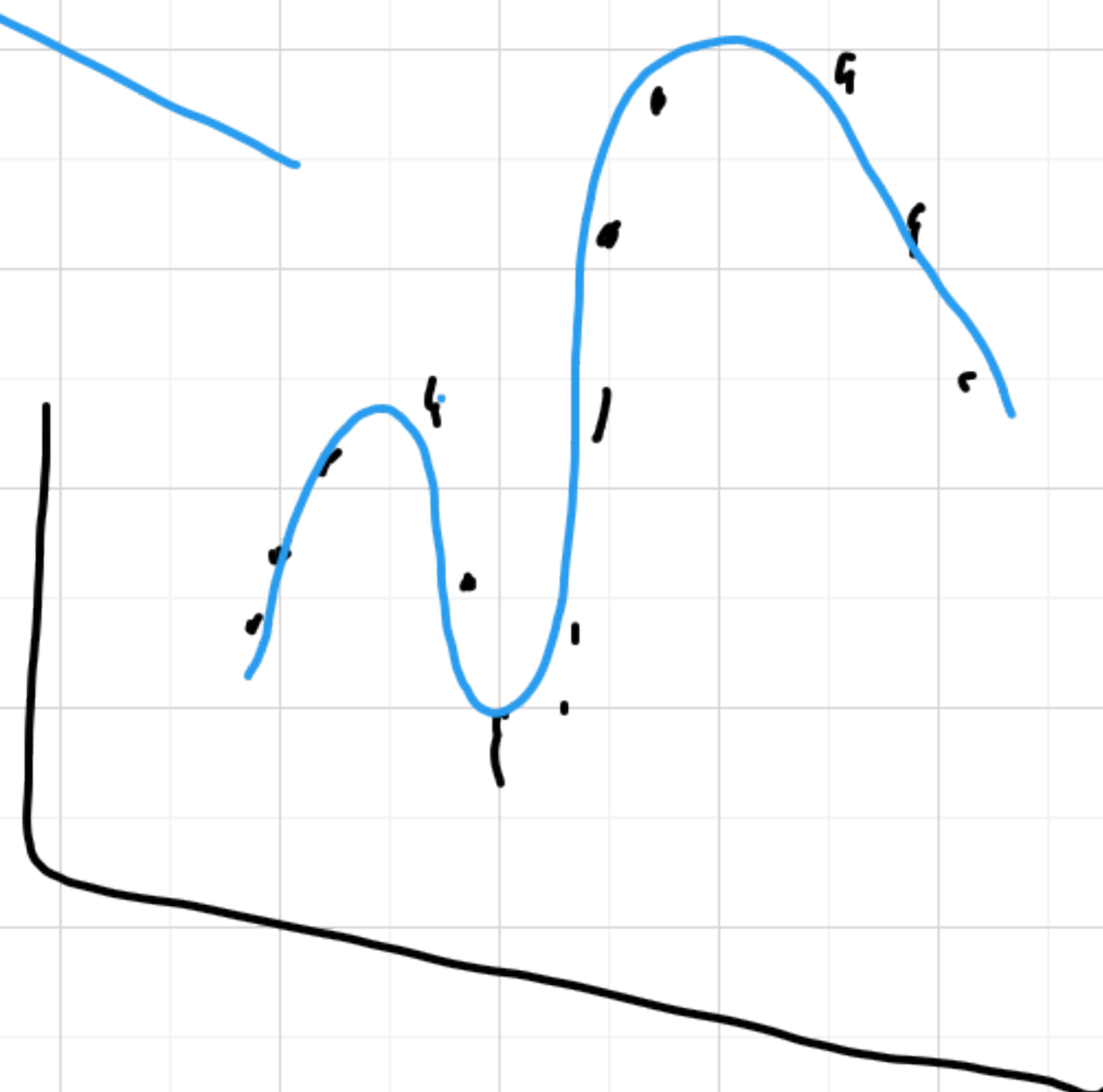
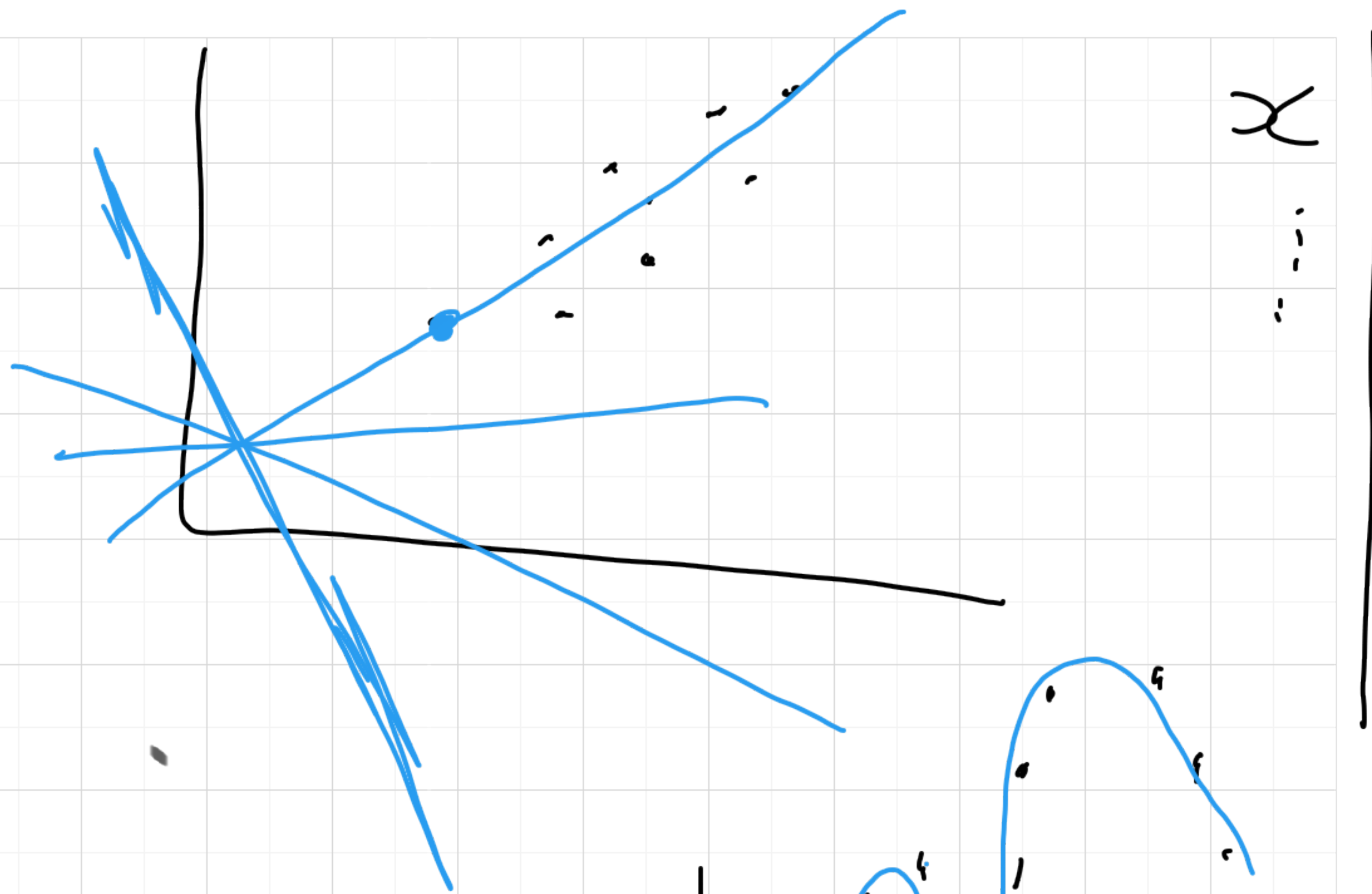
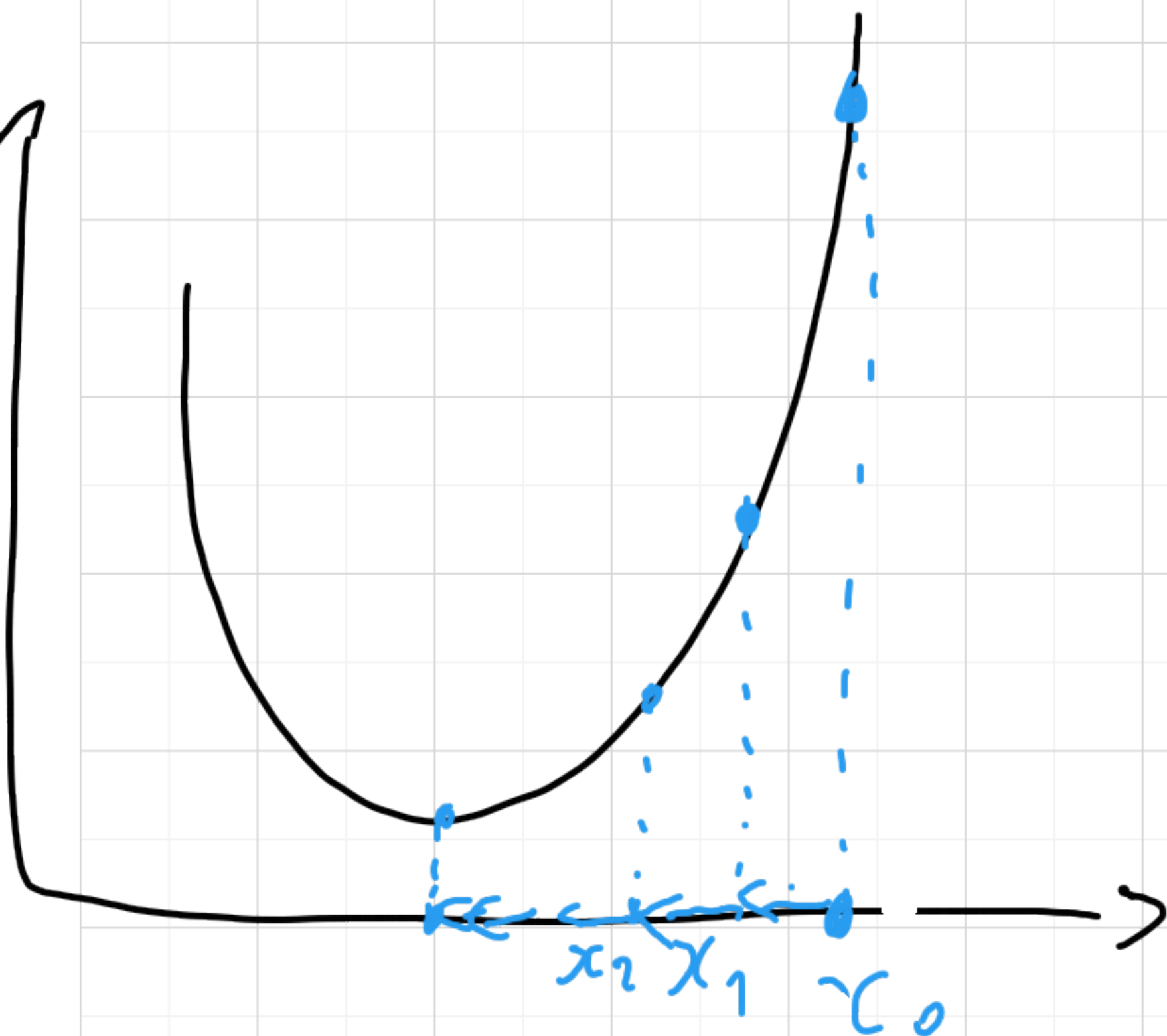
Back propagation



$$f'(x_0) > 0$$

$$f(x_0 + h) > f(x_0)$$





y
:
:

$$\hat{y} = w_1 \cdot x_1 + b_0$$

δ
3.2
3.0

δ
5

$$MSE(y, \hat{y}) = \frac{\sum_i^D (y_i - \hat{y}_i)^2}{D}$$

$w_1 \cdot x_i + b_0$

0/0

$$\frac{\partial \text{MSE}(y_1, \hat{y}_1)}{\partial w_1} = \frac{\partial (y_1 - \hat{y}_1)^2}{\partial w_1} = \frac{\partial}{\partial w_1} \left[(y_1 - w_1 \cdot x + b)^2 \right] = 2 \cdot (-x + b)$$

$$\frac{\partial}{\partial} u^m = m \cdot \frac{\partial u(x)}{\partial x} \cdot u^{m-1}$$

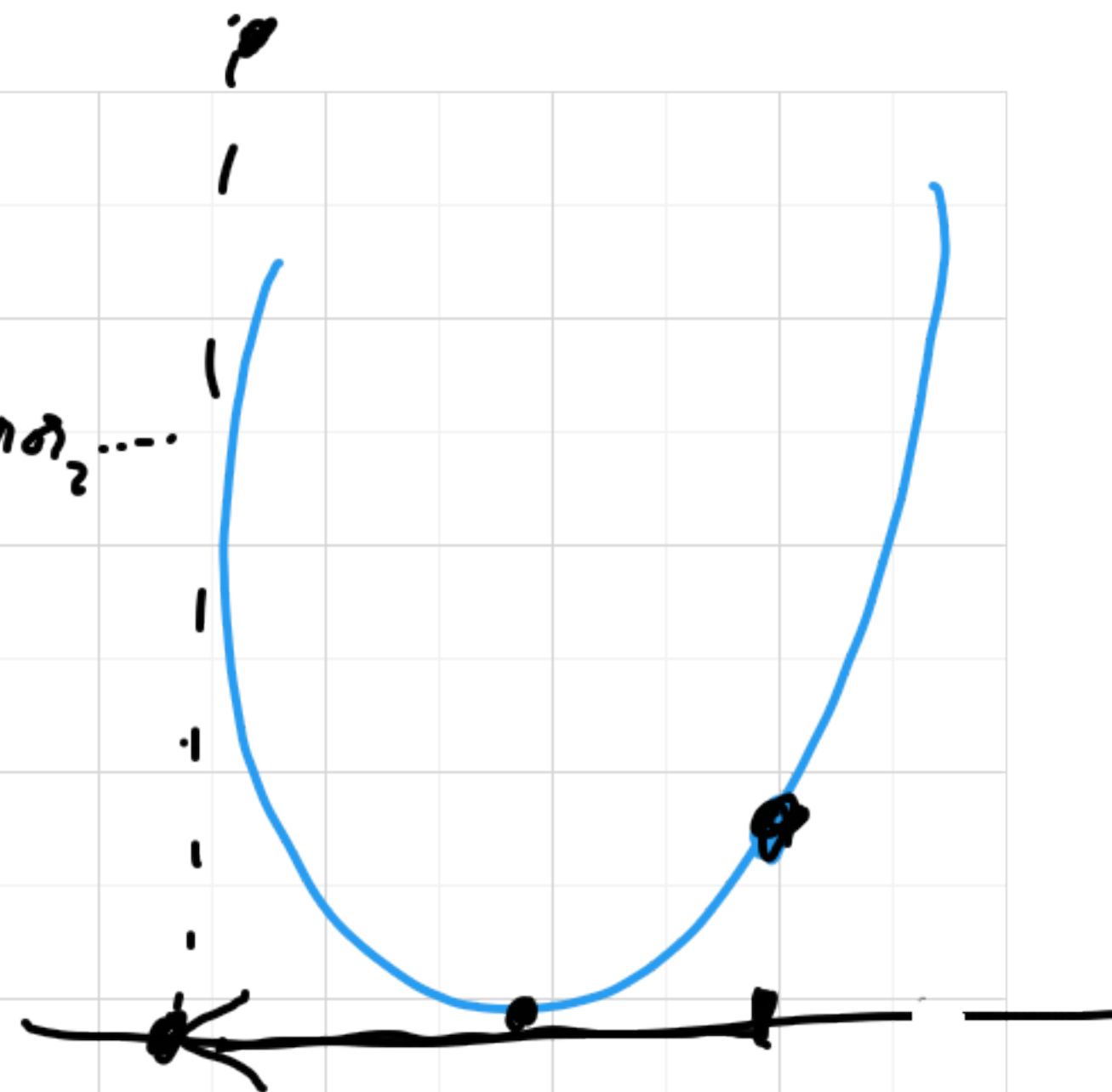
for all
 $\Rightarrow$

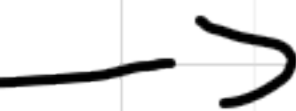
$$2 \cdot (-x_1) \cdot \text{error}_1 + 2 \cdot (-x_2) \cdot \text{error}_2 \dots$$

$x_c) \cdot \text{error}$

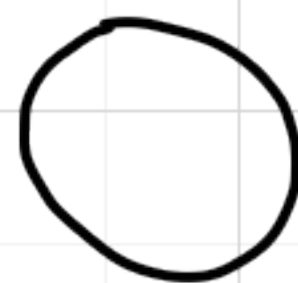
> 0
 < 0

$$w_{1\text{new}} = w_1 + \underset{\substack{\uparrow \\ 0.01}}{\eta} \cdot 2x \cdot \text{error}$$

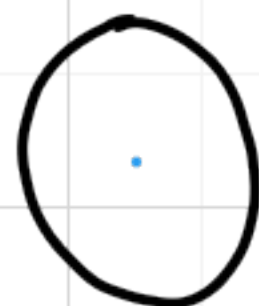




x



x'



x''



$$x''_1 = \frac{1}{1 + e^{-(w \cdot x')}} \quad \downarrow$$

$$\hat{y} = \frac{1}{1 + e^{-w \cdot x''}} \quad \uparrow$$

$$x'_1 = \frac{1}{1 + e^{-(w'_{11}x_1 + w'_{12}x_2 + b_0)}}$$

Computational Graphs

$$d = a \times b + c$$

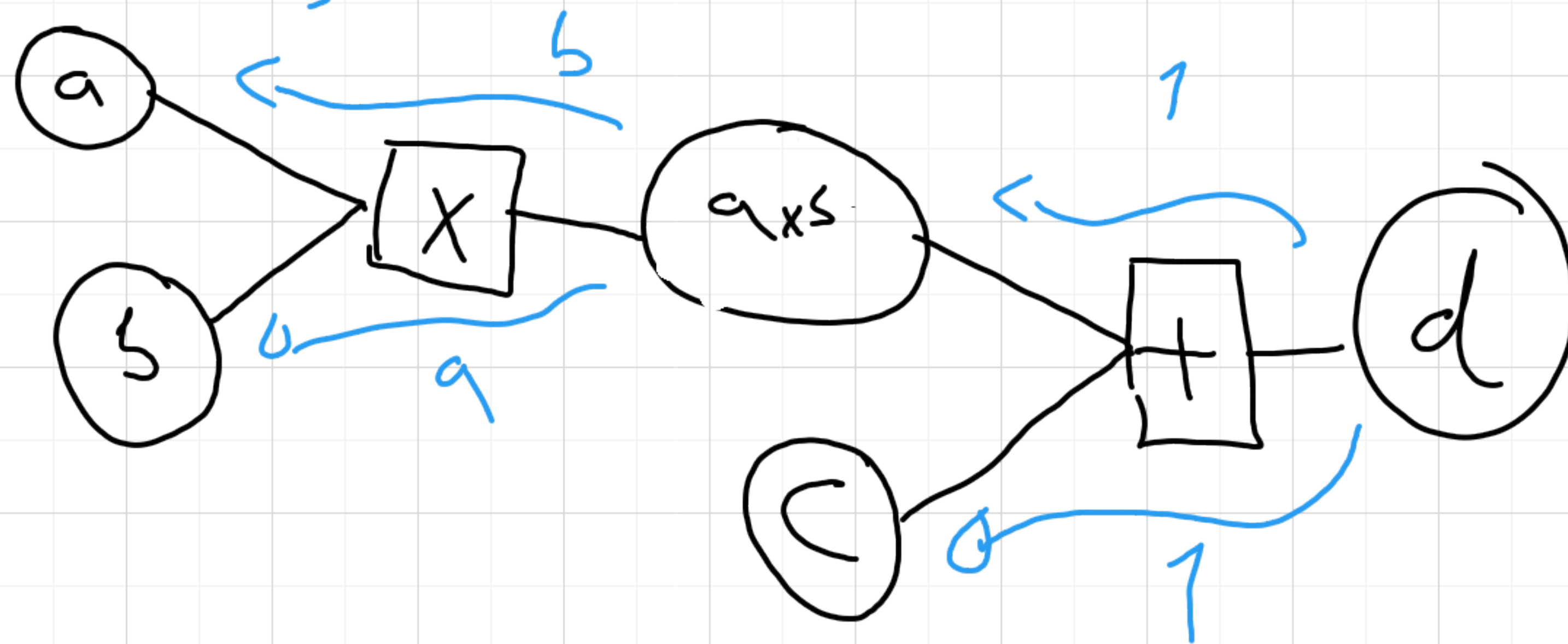
$$+1 \rightarrow 8$$

$$+2$$
$$+3$$

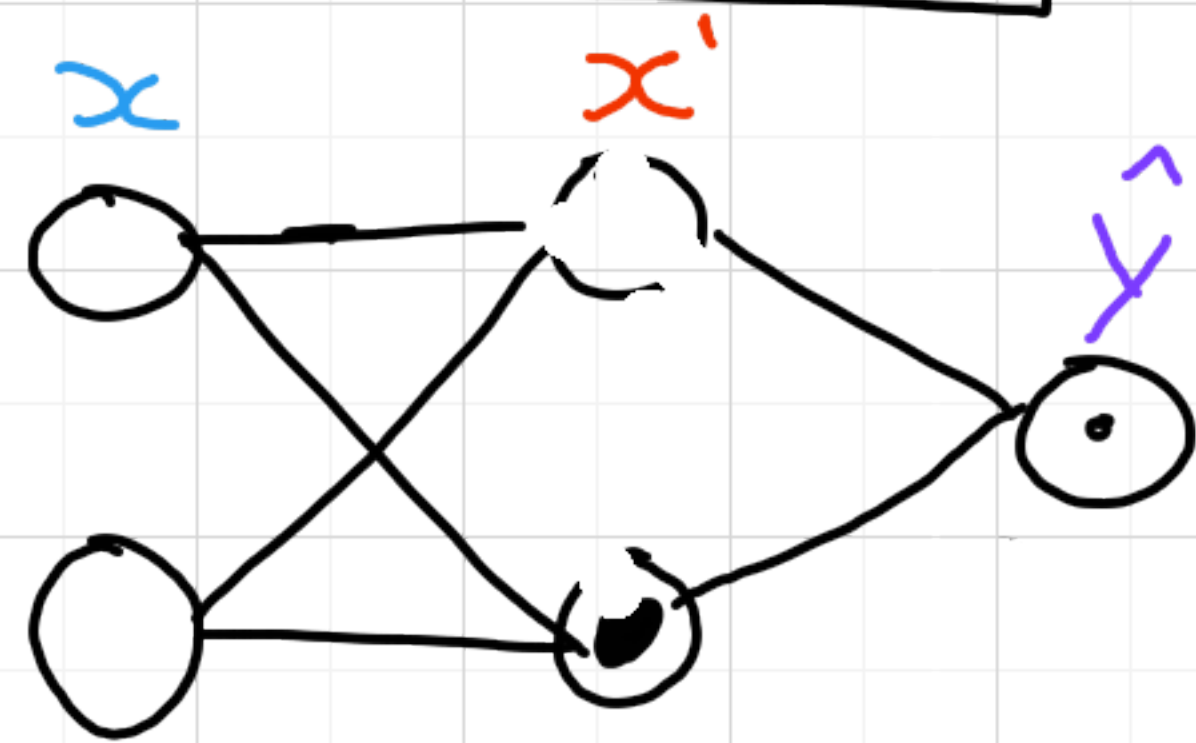
$$a \quad b$$
$$3 \times 4 = 12$$

$$3 \times (4+1) = 12 + 3$$

$$\frac{\partial d}{\partial a}$$



Exercice



$$x'_1 = \sigma(w_1)$$

$$x'_2 = \sigma(w_2)$$

$$\hat{y} = \sigma(w'_1 \cdot x'_1)$$

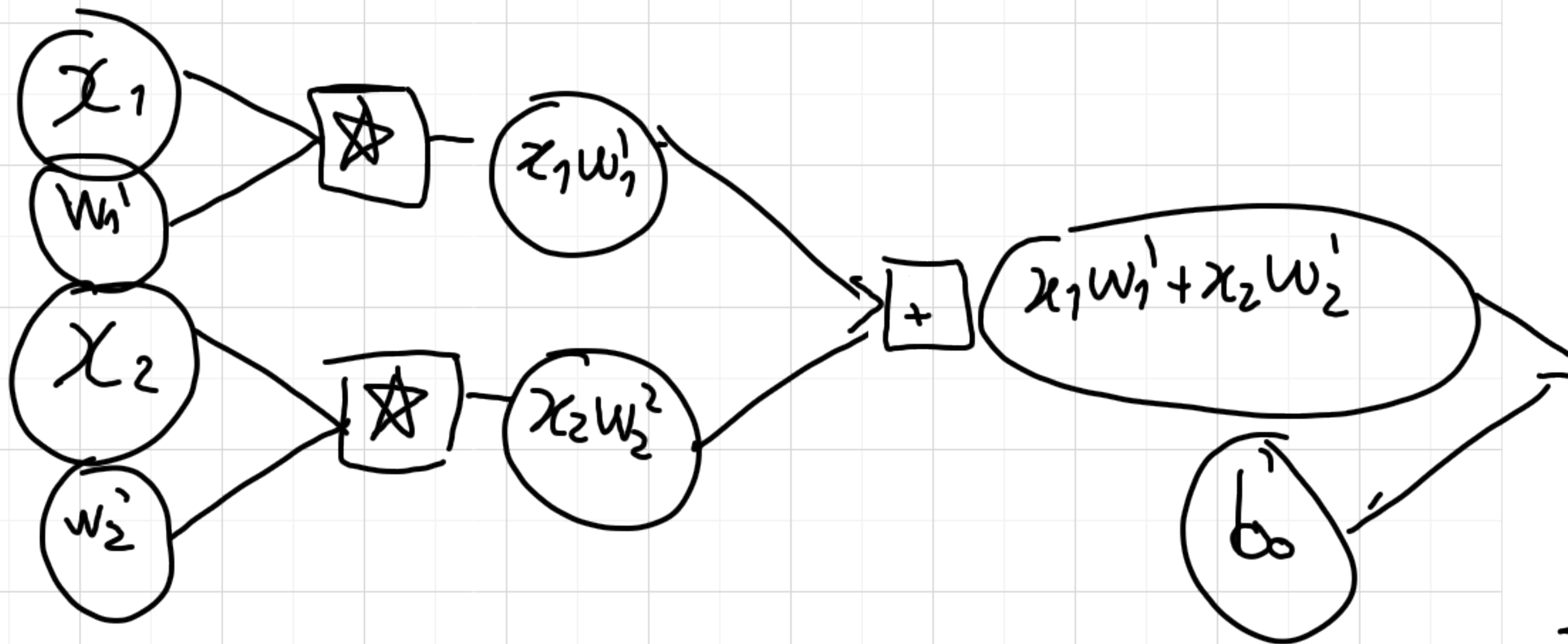
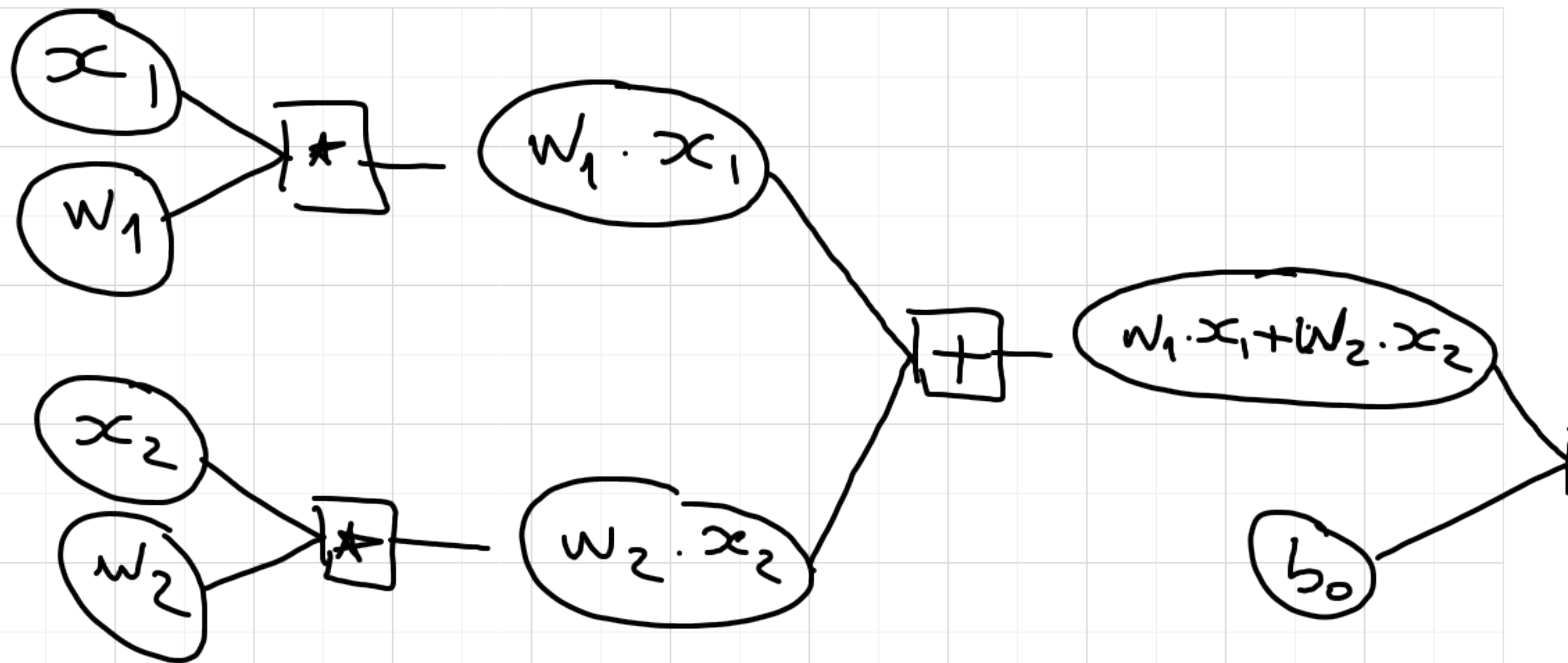
$$= \frac{\partial d}{\partial a \times b} \times \frac{\partial a \times b}{\partial a}$$

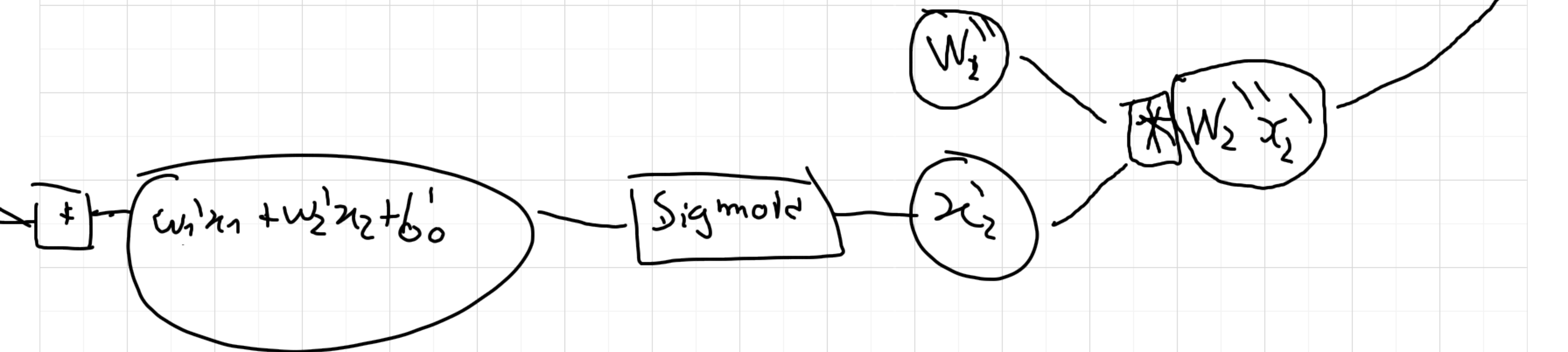
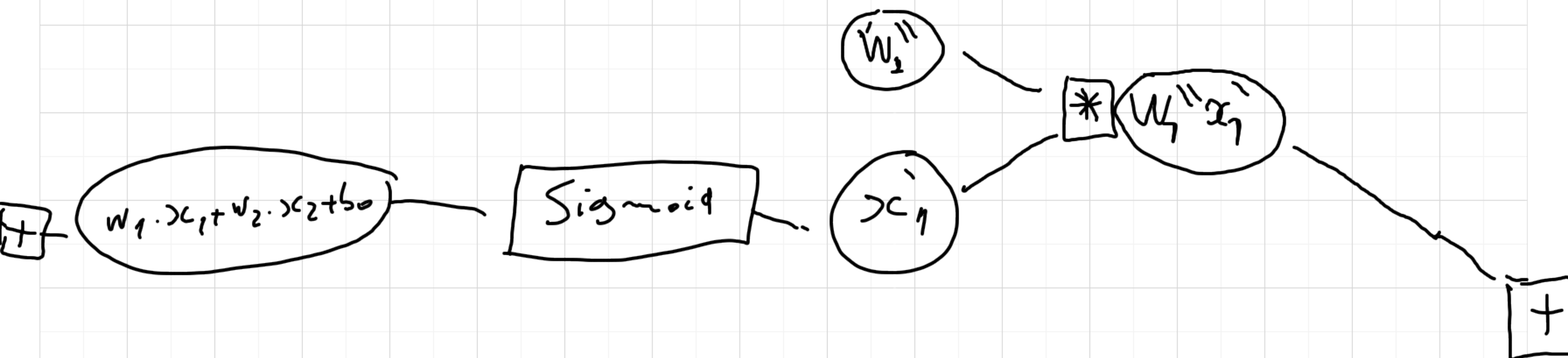
$$\frac{\partial d}{\partial a} = 1 \cdot b = b$$

$$w_1 \cdot x_1 + w_2 \cdot x_2 + b_0)$$

$$w'_1 \cdot x_1 + w'_2 \cdot x_2 + b'_0)$$

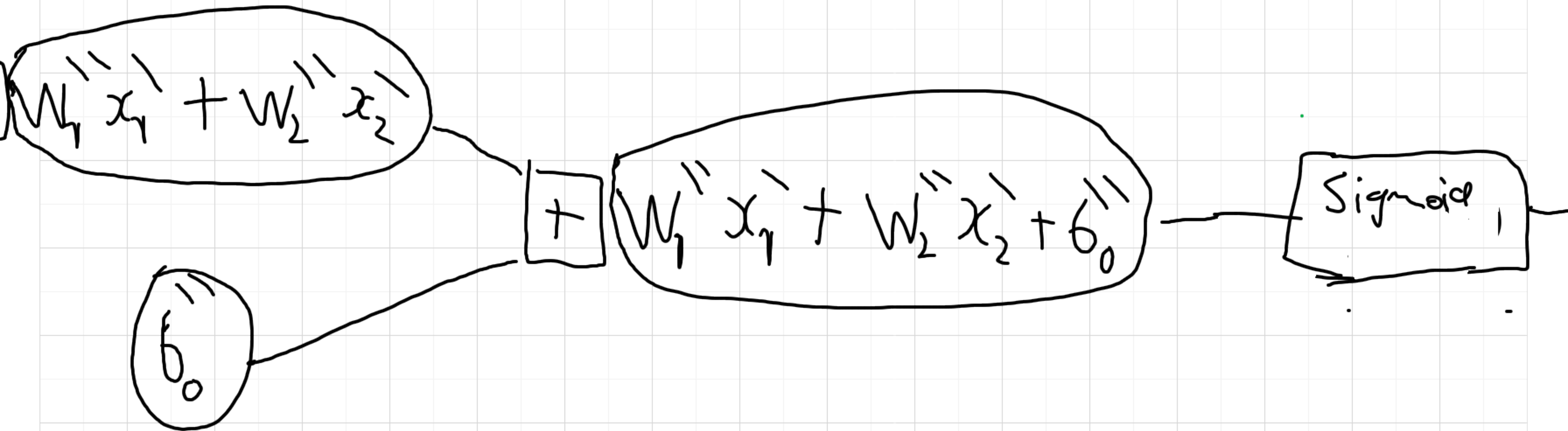
$$+ w''_2 x'_2 + b''_0)$$



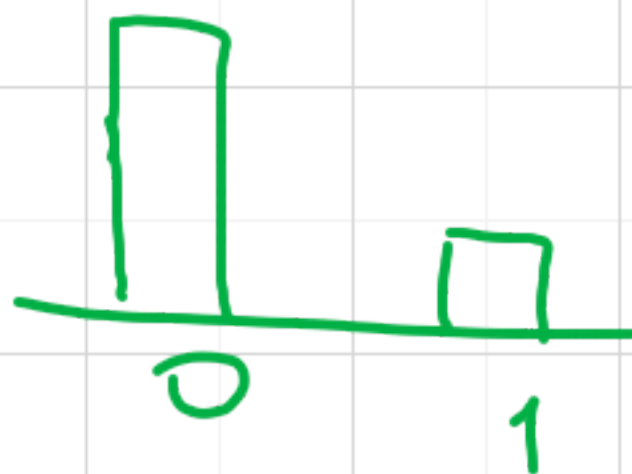
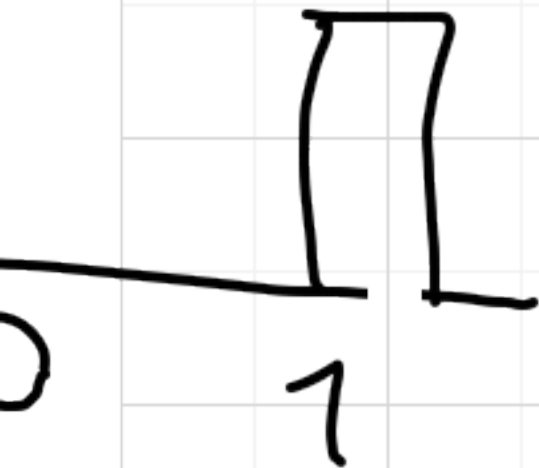


$$CE(y, \hat{y}) = y \left[\frac{1}{1 + e^{-\hat{y}}} \right]$$

① →



$$-\log \hat{y} + (1-y) \cdot \log(1-\hat{y})$$



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